

Inflation for the Classical Triangle: Nontermination and Quantitative Complexity

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Abstract

No finite order of classical inflation characterizes compatibility in the binary triangle. This answers a question of Navascués and Wolfe, whose hierarchy rejects each incompatible distribution at some finite order. For every order t , we construct a rational three-bit distribution that passes the full symmetric test with its tensor-power diagonal law, as well as the ancestral-independence and recursively expressible-set constraints, but violates the Finner inequality. The witness uses independent Bernoulli defects on a cube of source-copy indices. Along a second explicit family P_ε approaching a compatible point, the first rejecting order is $\Theta(\varepsilon^{-1/3})$, with asymptotic lower and upper constants $1/2$ and $4 - 2\sqrt{3}$. We obtain the upper bound from a family of finite-order inequalities and compute the leading total-variation and Euclidean distances to the compatible set. The rejecting order therefore grows as the inverse cube root of distance along this family, while another family approaches compatibility with rejecting order two. A second-moment inequality gives an explicit rejecting order for every Finner-violating law. For a general correlation scenario with L sources and K joint outcomes, a total-variation distance promise $\delta > 0$ gives the bound $\lfloor LK/(4\delta^2) \rfloor + 1$. For rational binary-triangle laws encoded in at most B bits, the worst required order is $2^{\Theta(B)}$. All proofs are analytic; exact finite certificates accompany the paper.

1 Introduction

In the classical triangle, three independent latent sources generate three observations through responses $A = A(X, Z)$, $B = B(X, Y)$ and $C = C(Z, Y)$. The sources may have arbitrary alphabets; here the observations are binary. Their compatible joint laws form a nonconvex set. Finite latent-cardinality bounds imply that this set has a finite semialgebraic description, but an explicit characterization remains unavailable even in the binary case [5, 9].

Wolfe, Spekkens and Fritz introduced inflation as a test of causal compatibility [11]. At order t , one copies each source t times and asks for a joint law on the copied observations with specified marginals. Navascués and Wolfe proved that their hierarchy is asymptotically complete [7]. They also asked whether a fixed scenario and fixed observed alphabets admit a finite characterizing order. We give a negative answer for the binary triangle and quantify the order needed along an explicit family approaching compatibility.

Theorem A: no finite order. For the triangle no finite order suffices, already for binary observed variables. For every integer $t \geq 1$ put

$$\varepsilon_t = \frac{1}{2t^3}, \quad r_t = (1 - \varepsilon_t)^{t-1}, \quad P_t = \varepsilon_t \delta_{000} + (1 - \varepsilon_t) \text{Bern}(r_t)^{\otimes 3},$$

where $\text{Bern}(r)$ takes the value 1 with probability r .

Theorem 1.1. *For every $t \geq 1$, P_t satisfies the complete order- t Navascués–Wolfe inflation test for the triangle and is not compatible with the triangle: $P_t(000)^2 - P_{t,A}(0)P_{t,B}(0)P_{t,C}(0) \geq \varepsilon_t^2/2 > 0$, in violation of the Finner inequality that every triangle-compatible distribution obeys. Consequently, for the triangle with observed alphabets of any sizes ≥ 2 , no finite order of the hierarchy characterizes classical compatibility.*

We use the full order- t test of [7]: a single normalized distribution on all $3t^2$ copied observations, invariant under independent permutations of the three families of copy indices, whose t diagonal copies of the triangle are jointly distributed as $P_t^{\otimes t}$. The global law we construct satisfies, in addition, every ancestral-independence product prescription and every recursively expressible-set prescription of [11], so the conclusion holds for those stronger finite tests as well.

The witness depends on the order: $\forall t \exists P_t$. Asymptotic completeness still rejects each fixed incompatible distribution at some finite order.

Theorem B: a sharp rejecting-order exponent. The second result measures how fast the required order grows along one family. For $0 < \varepsilon < 1/8$, put $P_\varepsilon = \varepsilon \delta_{000} + (1-\varepsilon) \text{Bern}(1 - \frac{1}{2}\varepsilon^{2/3})^{\otimes 3}$; these laws are incompatible and converge to the compatible point δ_{111} as $\varepsilon \downarrow 0$. Their first rejecting order satisfies

$$\frac{1}{2} \leq \liminf_{\varepsilon \downarrow 0} \varepsilon^{1/3} t_{\min}^H(P_\varepsilon) \leq \limsup_{\varepsilon \downarrow 0} \varepsilon^{1/3} t_{\min}^H(P_\varepsilon) \leq 4 - 2\sqrt{3}$$

for each of the three hierarchies (Proposition 5.1). We prove the exponent $1/3$; we do not prove that a limiting constant exists or what it is. The lower bound is the passing construction of Theorem A. The upper bound is an inequality that every order- t feasible law obeys, $tP(000) \leq P_A(0) + \binom{t}{2}P_B(0)P_C(0)$ (Theorem 4.1). A second-moment inequality in the same theorem gives an explicit rejecting order for any Finner-violating law. Theorem 6.1 shows that ε is, to leading order, $(1 - 2^{-3/2})^{-1}$ times the total-variation distance from P_ε to the compatible set, so along this family the rejecting order diverges like the inverse cube root of the distance to compatibility. The statement is about one direction of approach to the boundary; Proposition 4.2 gives another family approaching compatibility whose rejecting order is 2.

Bounds from distance and input size. For a correlation scenario with L sources and K joint outcomes, a promised total-variation distance $\delta > 0$ from compatibility gives the rejecting-order bound $\lfloor LK/(4\delta^2) \rfloor + 1$ (Proposition 7.1). This follows from the convergence argument of [7]. For rational binary-triangle laws with at most B bits, the worst required order is $2^{\Theta(B)}$ (Proposition 8.1). These are bounds on inflation order; the semialgebraic description also gives a separate decision procedure for exact compatibility.

Method. The construction is a “defect cube”: independent $\text{Bernoulli}(\varepsilon)$ bits D_{ijk} on the cube $[t]^3$, with $A^{ij} = \prod_k (1 - D_{ijk})$, $B^{jk} = \prod_i (1 - D_{ijk})$, $C^{ik} = \prod_j (1 - D_{ijk})$. Each copied triangle sees the three lines through one cell, which meet only at that cell, so its law is the mixture above; the t diagonal triangles use disjoint families of cells, so they are independent; and permuting copy indices permutes independent bits. Incompatibility is the Finner inequality $P(000)^2 \leq P_A(0)P_B(0)P_C(0)$, proved by two applications of Cauchy–Schwarz for arbitrary stochastic responses and latent alphabets, following [3, 8]. The computer-assisted content consists

of exact rational certificates at orders 1, 2 and 3 and for the fan inequality along P_ε ; they check the analytic claims on finite tables and the proofs do not use them.

2 The classical triangle and finite inflation tests

2.1 The triangle

Three mutually independent latent variables X, Y, Z take values in arbitrary measurable spaces with laws μ_X, μ_Y, μ_Z . Three observed variables with finite alphabets are generated by stochastic response kernels,

$$A \sim \kappa_A(\cdot \mid X, Z), \quad B \sim \kappa_B(\cdot \mid X, Y), \quad C \sim \kappa_C(\cdot \mid Z, Y),$$

conditionally independently given the sources. Write $\mathcal{C}_\Delta$ for the set of joint laws of (A, B, C) obtained this way for binary alphabets; the notation extends to any fixed finite alphabets. Assign each node an independent random seed and include that seed in one of its incident sources; the other node sharing that source ignores it. This preserves the triangle graph and shows that stochastic and deterministic responses generate the same set of observed laws [11]; we use whichever description is convenient. By a theorem of Rosset, Gisin and Wolfe [9], latent alphabets may be taken finite of bounded size, so $\mathcal{C}_\Delta$ is compact and semialgebraic; we use compactness in Section 6 and semialgebraicity in Section 8.

2.2 Inflation at order t

Fix $t \geq 1$. The order- t inflation of the triangle has t independent copies $X_1, \dots, X_t$ of X , and likewise of Y and Z , and the $3t^2$ copied observations

$$A^{ij} = A(X_i, Z_j), \quad B^{ik} = B(X_i, Y_k), \quad C^{jk} = C(Z_j, Y_k), \quad i, j, k \in [t],$$

each with the response kernel of its original. The copied ancestors are $\{X_i, Z_j\}, \{X_i, Y_k\}, \{Z_j, Y_k\}$. A *copied triangle* is $\Delta_{ijk} = \{A^{ij}, B^{ik}, C^{jk}\}$; the *diagonal triangles* are $\Delta_{\ell\ell\ell}, \ell \in [t]$. If $P \in \mathcal{C}_\Delta$ is realized by some model, then running the inflated model produces a joint law Γ_t on all copied observations with the following properties, which are therefore necessary for compatibility: independent permutations of the three index families leave Γ_t invariant, every copied triangle has law P , and the diagonal triangles, which depend on disjoint sets of source copies, are independent. The finite tests below ask for a joint law with some of these properties.

Definition 2.1 (Navascués–Wolfe feasible set). $P \in \mathcal{I}_t^{\text{NW}}$ if there is a probability law Γ_t on $\{0, 1\}^{3t^2}$ such that

- (i) Γ_t is invariant under independent permutations of the X -, Z - and Y -copy indices, that is under the group S_t^3 acting on the labels (i, j, k) ; and
- (ii) the diagonal law is a tensor power: $\text{Law}_{\Gamma_t}((A^{\ell\ell}, B^{\ell\ell}, C^{\ell\ell})_{\ell=1}^t) = P^{\otimes t}$.

This is the full-table symmetric formulation of [7] (equations (16)–(19) in the published numbering; the arXiv version numbers them (17)–(20)), whose degree- g diagonal conditions for $g \leq t$ follow from (ii) by marginalization. Navascués and Wolfe index the copies cyclically as $A(U_1^i, U_2^j), B(U_2^i, U_3^j), C(U_3^i, U_1^j)$; our labelling A^{ij}, B^{ik}, C^{jk} names the same variables with the shared index written in the same position, and the symmetry group is the same. They also note that further relations hold in a genuine inflation, such as $\text{Law}(A^{12}, B^{21}, C^{12}) = P_A \otimes P_B \otimes P_C$

at order two (written here in our indexing), and do not impose them in the hierarchy; those relations are instances of the ancestral-independence prescriptions of Definition 2.2 below, which our witness also satisfies. Membership in the Navascués–Wolfe feasible set is a linear feasibility problem in the 2^{3t^2} entries of Γ_t . Navascués and Wolfe prove that $\mathcal{I}_t^{\text{NW}} \downarrow \mathcal{C}_\Delta$ as $t \rightarrow \infty$, with a quantitative rate recalled in Section 7.

Definition 2.2 (Injectable sets and the AI feasible set). A set of copied observations is *injectable* if erasing copy indices maps it injectively onto a subset of $\{A, B, C\}$ and the shared latent indices agree; for the triangle these are the subsets of copied triangles. Two sets are *ancestrally independent* if their copied latent ancestors are disjoint. $P \in \mathcal{I}_t^{\text{AI}}$ if $P \in \mathcal{I}_t^{\text{NW}}$ with a witness Γ_t under which every injectable set has the corresponding marginal of P and every union of pairwise ancestrally independent injectable sets has the product of the corresponding marginals.

Definition 2.3 (Recursively expressible feasible set). Following [11, Definition 7], expressible sets are generated from injectable sets by two rules: if X' and Y' are d -separated by Z' in the inflated causal graph and the laws on $X' \cup Z'$ and $Y' \cup Z'$ are prescribed, then $p(x', y', z') = p(x', z')p(y', z')/p(z')$ (and 0 when $p(z') = 0$) is prescribed on $X' \cup Y' \cup Z'$; and marginals of prescribed sets are prescribed. $P \in \mathcal{I}_t^{\text{exp}}$ if $P \in \mathcal{I}_t^{\text{AI}}$ with a witness satisfying every prescription so generated.

For fixed P , each prescription fixes a marginal of Γ_t , so the strengthened tests are also linear feasibility problems. Restricting a witness to fewer copy indices preserves all three sets of conditions. Thus each hierarchy decreases with t , and at a fixed order the feasible sets are nested,

$$\mathcal{I}_t^{\text{exp}} \subseteq \mathcal{I}_t^{\text{AI}} \subseteq \mathcal{I}_t^{\text{NW}}, \quad (1)$$

and a test is stronger when its feasible set is smaller. All three contain $\mathcal{C}_\Delta$. We prove our witness lies in the smallest set, so the conclusion holds for all three; the Navascués–Wolfe set is the one whose asymptotic completeness is published and the one named in the question we answer.

For $H \in \{\text{NW}, \text{AI}, \text{exp}\}$ and $P \notin \mathcal{C}_\Delta$, write

$$t_{\min}^H(P) = \min\{t \geq 1 : P \notin \mathcal{I}_t^H\}.$$

Asymptotic completeness of the NW hierarchy and (1) make this a finite integer. We use

$$d_{\text{TV}}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|, \quad d_{\text{TV}}(P, \mathcal{C}) = \inf_{Q \in \mathcal{C}} d_{\text{TV}}(P, Q),$$

and write $d_2(P, \mathcal{C}) = \inf_{Q \in \mathcal{C}} \|P - Q\|_2$ for Euclidean distance.

2.3 The Finner inequality

Every $P \in \mathcal{C}_\Delta$ satisfies

$$P(000)^2 \leq P_A(0) P_B(0) P_C(0), \quad (2)$$

where P_A, P_B, P_C are the one-variable marginals. This is the event form of Finner’s inequality [3], which Renou et al. [8] applied to the triangle network; we include a short proof in Section 3.2 because we need it for arbitrary stochastic responses and arbitrary latent alphabets, and because the incompatibility argument rests on it.

3 The explicit nontermination family

For $0 < \varepsilon < 1$ and $0 \leq r \leq 1$ define the three-bit law

$$Q(\varepsilon, r) = \varepsilon \delta_{000} + (1 - \varepsilon) \text{Bern}(r)^{\otimes 3}, \quad (3)$$

a point mass at 000 with weight ε mixed with three independent bits that equal 1 with probability r . Throughout, $\text{Bern}(r)$ denotes the law on $\{0, 1\}$ with $\text{Bern}(r)(1) = r$.

Theorem 3.1 (Membership at every finite order). *Let $t \geq 1$, $0 < \varepsilon < 1$ and $0 \leq r \leq (1 - \varepsilon)^{t-1}$. Then*

$$Q(\varepsilon, r) \in \mathcal{I}_t^{\text{exp}} \subseteq \mathcal{I}_t^{\text{AI}} \subseteq \mathcal{I}_t^{\text{NW}}.$$

Theorem 3.2 (No finite characterizing order). *For every integer $t \geq 1$, with $\varepsilon_t = 1/(2t^3)$ and $r_t = (1 - \varepsilon_t)^{t-1}$, the law $P_t = Q(\varepsilon_t, r_t)$ satisfies*

$$P_t \in \mathcal{I}_t^{\text{exp}} \subseteq \mathcal{I}_t^{\text{AI}} \subseteq \mathcal{I}_t^{\text{NW}} \quad \text{and} \quad P_t(000)^2 - P_{t,A}(0)P_{t,B}(0)P_{t,C}(0) \geq \frac{\varepsilon_t^2}{2} > 0,$$

so $P_t \notin \mathcal{C}_\Delta$. Hence none of the three finite-order tests characterizes $\mathcal{C}_\Delta$ at any finite order.

Theorem 3.2 includes $t = 1$, where $P_1 = \frac{1}{2}(\delta_{000} + \delta_{111})$, the perfectly correlated three-bit law, which is the standard example of a triangle-incompatible distribution and passes the trivial order-one test.

Corollary 3.3 (Larger alphabets). *For the triangle with observed alphabets of any sizes $n_A, n_B, n_C \geq 2$, no finite order of the Navascués–Wolfe hierarchy characterizes the compatible set.*

Proof. Regard P_t as a law on the larger alphabets, supported on $\{0, 1\}^3$. A compatible realization of a law supported on $\{0, 1\}^3$ has response kernels that output values outside $\{0, 1\}$ with probability zero, so it is a binary realization; hence P_t remains incompatible. The order- t witness Γ_t of Theorem 3.2, viewed as a law on the larger copied alphabets, still has the required symmetry and diagonal law. So $P_t \in \mathcal{I}_t^{\text{NW}} \setminus \mathcal{C}$ for the larger scenario as well. $\square$

3.1 The defect cube

Fix $t \geq 1$, $0 < \varepsilon < 1$ and $0 \leq r \leq (1 - \varepsilon)^{t-1}$. The witness is a joint law Γ_t on the $3t^2$ copied observations built from independent bits indexed by the cells of the cube $[t]^3$.

Roots. Let $D_{ijk} \sim \text{Bern}(\varepsilon)$, $(i, j, k) \in [t]^3$, be mutually independent *defect* bits, and let $N_v \sim \text{Bern}(s)$, one for each copied observation v , be mutually independent *private* bits, independent of the defects, with

$$s = 1 - \frac{r}{(1 - \varepsilon)^{t-1}} \in [0, 1]. \quad (4)$$

The endpoint $r = 0$ uses $s = 1$; the endpoint $r = (1 - \varepsilon)^{t-1}$ uses $s = 0$, in which case the private bits play no role.

Outputs. Define

$$\begin{aligned} A^{ij} &= \mathbf{1}\{N_{A^{ij}} = 0 \text{ and } D_{ijk} = 0 \forall k\}, \\ B^{ik} &= \mathbf{1}\{N_{B^{ik}} = 0 \text{ and } D_{ijk} = 0 \forall j\}, \\ C^{jk} &= \mathbf{1}\{N_{C^{jk}} = 0 \text{ and } D_{ijk} = 0 \forall i\}. \end{aligned} \tag{5}$$

So A^{ij} reads the line of cells $\Lambda(A^{ij}) = \{(i, j, k) : k \in [t]\}$, B^{ik} the line $\Lambda(B^{ik}) = \{(i, j, k) : j \in [t]\}$, and C^{jk} the line $\Lambda(C^{jk}) = \{(i, j, k) : i \in [t]\}$; each output equals 1 exactly when its private bit and every defect on its line are 0. Let Γ_t be the joint law of the outputs. For $s = 0$ its entries are

$$\Gamma_t(w) = \sum_{d \in \{0,1\}^{[t]^3} : F_t(d)=w} \varepsilon^{|d|} (1 - \varepsilon)^{t^3 - |d|}, \tag{6}$$

For rational ε , these entries are rational. Here F_t is the map (5) from defect configurations to observed assignments; they are nonnegative and sum to $(\varepsilon + 1 - \varepsilon)^{t^3} = 1$.

The defect cube is not a triangle model, and Γ_t does not come from running an inflated triangle model; it is a probability law on the copied observations, which is what a finite inflation test asks for.

Lemma 3.4 (Disjoint ancestry gives disjoint inputs). *If two sets of copied observations have disjoint copied latent ancestry in the order- t inflation, then their defect lines and their private bits are disjoint, and the two sets of outputs are independent under Γ_t .*

Proof. Two lines through the cube intersect only if they share the index that corresponds to a common source type: $\Lambda(A^{ij}) \cap \Lambda(B^{i'k}) \neq \emptyset$ forces $i = i'$, and then X_i is a common ancestor; similarly $\Lambda(A^{ij}) \cap \Lambda(C^{j'k}) \neq \emptyset$ forces $j = j'$ and $\Lambda(B^{ik}) \cap \Lambda(C^{j'k'}) \neq \emptyset$ forces $k = k'$; two lines of the same type are disjoint unless the observations coincide. Taking unions, disjoint ancestry implies disjoint defect supports. Distinct observations have distinct private bits. Outputs that are functions of disjoint families of independent bits are independent. $\square$

Lemma 3.5 (Law of a copied triangle). *For every $(i, j, k) \in [t]^3$, $\text{Law}_{\Gamma_t}(A^{ij}, B^{ik}, C^{jk}) = Q(\varepsilon, r)$.*

Proof. The three lines of Δ_{ijk} meet pairwise exactly at the cell (i, j, k) . Condition on D_{ijk} . If $D_{ijk} = 1$, all three outputs are 0. If $D_{ijk} = 0$, each output depends on its own private bit and on the $t - 1$ remaining cells of its line; these three families are pairwise disjoint, so the outputs are conditionally independent, each equal to 1 with probability $(1 - s)(1 - \varepsilon)^{t-1} = r$ by (4). Averaging over D_{ijk} gives (3). $\square$

Lemma 3.6 (Symmetry). *Γ_t is invariant under independent permutations of the X -, Z - and Y -copy indices.*

Proof. A permutation σ of the X -indices sends $D_{ijk} \mapsto D_{\sigma(i)jk}$, $N_{A^{ij}} \mapsto N_{A^{\sigma(i)j}}$, and so on; it permutes the two families of i.i.d. bits and carries (5) to itself. The same holds for the other two index families. $\square$

3.2 Genuine incompatibility

Lemma 3.7 (Finner inequality for the triangle). *Every $P \in \mathcal{C}_\Delta$ satisfies the inequality $P(000)^2 \leq P_A(0)P_B(0)P_C(0)$.*

Proof. Let the sources have laws μ_X, μ_Y, μ_Z on arbitrary measurable spaces and let

$$f(x, z) = \Pr(A = 0 \mid x, z), \quad g(x, y) = \Pr(B = 0 \mid x, y), \quad h(z, y) = \Pr(C = 0 \mid z, y)$$

be the response probabilities, measurable functions with values in $[0, 1]$. Conditional independence of the responses given the sources gives $P(000) = \int f(x, z)g(x, y)h(z, y) d\mu_X d\mu_Y d\mu_Z$. Put $u(x)^2 = \int g(x, y)^2 d\mu_Y(y)$ and $v(z)^2 = \int h(z, y)^2 d\mu_Y(y)$. Cauchy–Schwarz in y , then Cauchy–Schwarz in the independent pair (x, z) , then $f^2 \leq f$, $g^2 \leq g$, $h^2 \leq h$ (all three functions lie in $[0, 1]$) give

$$\begin{aligned} P(000) &\leq \int f(x, z)u(x)v(z) d\mu_X d\mu_Z \\ &\leq \left(\int f^2 d\mu_X d\mu_Z \right)^{1/2} \left(\int u^2 d\mu_X \int v^2 d\mu_Z \right)^{1/2} \leq \sqrt{P_A(0)P_B(0)P_C(0)}. \end{aligned}$$

Tonelli’s theorem justifies every interchange, since all integrands are nonnegative and bounded. The argument uses neither finiteness of the latent alphabets nor determinism of the responses. $\square$

The inequality is the triangle case of Finner’s generalized Hölder inequality [3]; Renou et al. [8] derived it for the triangle network in this event form and proved it for quantum sources and for no-signalling boxes combined by local wirings.

Lemma 3.8 (Violation by the witness). *Let $t \geq 1$, $0 < \varepsilon < t^{-3}$ and $P = Q(\varepsilon, (1 - \varepsilon)^{t-1})$. Then $P(000)^2 > P_A(0)P_B(0)P_C(0)$; in particular $P \notin \mathcal{C}_\Delta$. For $\varepsilon = \varepsilon_t = 1/(2t^3)$,*

$$P_t(000)^2 - P_{t,A}(0)P_{t,B}(0)P_{t,C}(0) \geq \varepsilon_t^2 - t^3 \varepsilon_t^3 = \frac{\varepsilon_t^2}{2}.$$

Proof. With $r = (1 - \varepsilon)^{t-1}$, $P(000) = \varepsilon + (1 - \varepsilon)(1 - r)^3 \geq \varepsilon$ and, by symmetry of the three coordinates, $P_A(0) = P_B(0) = P_C(0) = 1 - (1 - \varepsilon)r = 1 - (1 - \varepsilon)^t \leq t\varepsilon$ by Bernoulli’s inequality. Hence $P(000)^2 \geq \varepsilon^2$ and $P_A(0)P_B(0)P_C(0) \leq t^3 \varepsilon^3$, and $\varepsilon^2 > t^3 \varepsilon^3$ exactly when $\varepsilon < t^{-3}$. For $\varepsilon_t = 1/(2t^3)$ the difference is at least $\varepsilon_t^2(1 - t^3 \varepsilon_t) = \varepsilon_t^2/2$. $\square$

At $t = 1$ the witness is $P_1 = \frac{1}{2}(\delta_{000} + \delta_{111})$ with $P_1(000)^2 = 1/4$ and $P_{1,A}(0)^3 = 1/8$; the margin $1/8$ is $\varepsilon_1^2/2$.

3.3 Survival of every finite order

The defect law Γ_t witnesses $Q(\varepsilon, r) \in \mathcal{I}_t^{\text{exp}}$, which is Theorem 3.1. The three feasible sets require successively more of Γ_t ; we verify them in order.

3.3.1 The Navascués–Wolfe conditions

Symmetry is Lemma 3.6. For the diagonal law, let R_ℓ be the union of the three lines of the diagonal triangle $\Delta_{\ell\ell\ell}$:

$$R_\ell = \{(\ell, \ell, k) : k \in [t]\} \cup \{(\ell, j, \ell) : j \in [t]\} \cup \{(i, \ell, \ell) : i \in [t]\}. \quad (7)$$

Equivalently, R_ℓ is the set of cells with at least two coordinates equal to ℓ ; it has $3t - 2$ cells.

Lemma 3.9 (Diagonal supports are mutually disjoint). *If $\ell \neq m$ then $R_\ell \cap R_m = \emptyset$. Consequently the t diagonal triangles depend on mutually disjoint families of independent bits, are mutually independent under Γ_t , and their joint law is $Q(\varepsilon, r)^{\otimes t}$.*

Proof. A cell has three coordinates; it cannot have two of them equal to ℓ and two equal to $m \neq \ell$. The private bits of distinct diagonal triangles are distinct. Mutual independence of functions of disjoint families of independent variables follows, and Lemma 3.5 identifies each factor. $\square$

The lemma proves the tensor-power diagonal law, so $Q(\varepsilon, r) \in \mathcal{I}_t^{\text{NW}}$.

3.3.2 The ancestral-independence conditions

Every injectable set is a subset of some copied triangle Δ_{ijk} (Definition 2.2), so its prescribed marginal is a projection of $Q(\varepsilon, r)$ and is matched by Lemma 3.5. A union of pairwise ancestrally independent injectable sets is, by Lemma 3.4 applied to the union, a family of outputs depending on pairwise disjoint families of independent bits, hence mutually independent, with the prescribed product law. This proves $Q(\varepsilon, r) \in \mathcal{I}_t^{\text{AI}}$. In particular the higher-degree relations that Navascués and Wolfe mention but do not impose, such as $\text{Law}(A^{12}, B^{21}, C^{12}) = P_A \otimes P_B \otimes P_C$, hold for Γ_t : the three observations A^{12}, B^{21}, C^{12} have ancestor sets $\{X_1, Z_2\}$, $\{X_2, Y_1\}$ and $\{Z_1, Y_2\}$. Their lines $(1, 2, \cdot)$, $(2, \cdot, 1)$ and $(\cdot, 1, 2)$ are pairwise disjoint.

3.3.3 The recursively expressible conditions

The AI conditions do not by themselves give the conditional-product prescriptions of Definition 2.3, which use d -separations in the genuine inflated graph. Every such separation also holds for the defect law.

Lemma 3.10 (Latent-projection containment). *Let H_t be the directed acyclic graph with the defect and private bits as root vertices and the copied observations as their children according to (5). The latent projection of H_t onto the observed vertices is a subgraph of the latent projection of the order- t inflated triangle onto the same vertices.*

Proof. Both projections have only the observed vertices, all of which are sinks, and a bidirected edge between two observed vertices exactly when they share a latent parent. In H_t a private bit has one child and creates no edge, and a defect D_{ijk} has children A^{ij}, B^{ik}, C^{jk} , creating the three edges $A^{ij} \leftrightarrow B^{ik}$, $A^{ij} \leftrightarrow C^{jk}$, $B^{ik} \leftrightarrow C^{jk}$. In the genuine inflation these three pairs share X_i, Z_j and Y_k respectively, so the same three edges are present. $\square$

Lemma 3.11 (Recursive prescriptions hold). *Every marginal prescribed by the recursion of Definition 2.3 equals the corresponding marginal of Γ_t .*

Proof. Γ_t is Markov with respect to H_t , so every conditional independence among observed sets that is implied by d -separation in H_t holds under Γ_t . For these two-layer graphs, an active path between observed vertices alternates between latent roots and observed colliders; every internal observed vertex must belong to the conditioning set, since observed vertices have no descendants. Equivalently, in the bidirected projection an active path has all its internal vertices in the conditioning set. Deleting bidirected edges cannot create such a path. Lemma 3.10 therefore transfers every observed d -separation of the genuine inflation to H_t . Now induct on the recursion. Injectable base sets are matched by Lemma 3.5. At a conditional-product step with $X' \perp_d Y' \mid Z'$, the transferred conditional independence gives $\Gamma_t(x', y', z') = \Gamma_t(x', z')\Gamma_t(y', z')/\Gamma_t(z')$ whenever

$\Gamma_t(z') > 0$, while for $\Gamma_t(z') = 0$ both numerator marginals and the joint atom vanish, matching the zero branch. Marginalization of a matched set is matched. Copy-index equalities among prescribed marginals follow from Lemma 3.6. $\square$

This proves $Q(\varepsilon, r) \in \mathcal{I}_t^{\text{exp}}$ and completes the proof of Theorem 3.1. Theorem 3.2 follows by combining Theorem 3.1 at $r = r_t = (1 - \varepsilon_t)^{t-1}$ with Lemma 3.8.

4 Finite-order fan inequalities

The inequalities of this section bound, for every law feasible at order t , the probability of the all-zero atom by the one-variable zero marginals. They follow from the symmetry and the diagonal conditions of degrees one and two only.

Theorem 4.1 (Fan inequalities). *Let P be a law on three bits with $a = P_A(0)$, $b = P_B(0)$, $c = P_C(0)$, $z = P(000)$. If $P \in \mathcal{I}_t^{\text{NW}}$ then*

$$tz \leq a + \binom{t}{2}bc \quad \text{and} \quad t(z^2 - abc) \leq az - abc. \quad (8)$$

Consequently, if $z^2 > abc$ (a Finner violation), then

$$t_{\min}^{\text{NW}}(P) \leq \left\lfloor \frac{z \min\{a, b, c\} - abc}{z^2 - abc} \right\rfloor + 1, \quad (9)$$

and the same bound holds for $t_{\min}^{\text{AI}}$ and $t_{\min}^{\text{exp}}$.

This gives an explicit rejecting order in terms of four probabilities. Relabelling the local outcomes gives the corresponding bound for a Finner violation at any atom.

Proposition 4.2 (No uniformly divergent distance lower bound). *For $0 < p < 1$ let $R_p = (1 - p)\delta_{111} + p\delta_{000}$. Then $R_p \notin \mathcal{C}_{\Delta}$, $R_p \rightarrow \delta_{111}$ as $p \downarrow 0$, and $t_{\min}^H(R_p) = 2$ for all three hierarchies and all p . Hence no inequality $t_{\min}^H(P) \geq c d_{\text{TV}}(P, \mathcal{C}_{\Delta})^{-\alpha}$ with $c, \alpha > 0$ holds for all incompatible P ; the growth in Proposition 5.1 is a property of the direction of approach to the boundary.*

Proof of Theorem 4.1. For $t = 1$, the two inequalities reduce to $z \leq a$ and $z^2 \leq az$, which hold since $0 \leq z \leq a$. Now let Γ be an order- t feasible law for P , $t \geq 2$, and consider the $2t + 1$ copied observations A^{11}, B^{1k}, C^{1k} , $k \in [t]$: the fan of copied triangles $\Delta_{11k} = \{A^{11}, B^{1k}, C^{1k}\}$ sharing the vertex A^{11} . Define the indicators $A = \mathbf{1}[A^{11} = 0]$, $B_k = \mathbf{1}[B^{1k} = 0]$, $C_k = \mathbf{1}[C^{1k} = 0]$ and $T_k = AB_kC_k$. Three identities follow from Definition 2.1: $\mathbb{E}A = a$ and $\mathbb{E}T_k = z$, because a permutation of the Y -indices carries Δ_{11k} to the diagonal triangle Δ_{111} , whose law is P ; and for $k \neq \ell$,

$$\mathbb{E}[B_kC_{\ell}] = bc, \quad (10)$$

because the independent permutations $X_1 \mapsto 1$, $Z_1 \mapsto 2$, $Y_k \mapsto 1$, $Y_{\ell} \mapsto 2$ (extended arbitrarily to $[t]$) carry $B^{1k}, C^{1\ell}$ to B^{11}, C^{22} , which lie in the two diagonal triangles $\Delta_{111}, \Delta_{222}$ whose joint law is $P \otimes P$ by the degree-two diagonal condition. The argument uses no product prescription beyond the diagonal law, and (10) is not available for $k = \ell$.

First inequality. For every deterministic assignment of the fan,

$$\sum_k AB_kC_k \leq A + \frac{1}{2} \sum_{k \neq \ell} B_kC_{\ell}. \quad (11)$$

If $A = 0$ the left side vanishes. If $A = 1$, put $S = \sum_k B_k C_k$; then $\sum_k B_k \geq S$ and $\sum_k C_k \geq S$, so the right side is $1 + \frac{1}{2}[(\sum_k B_k)(\sum_k C_k) - S] \geq 1 + \frac{1}{2}(S^2 - S)$, and the right side minus the left side is at least $\frac{1}{2}(S - 1)(S - 2) \geq 0$ for every integer $S \geq 0$. Taking expectations under Γ and using the three identities gives $tz \leq a + \binom{t}{2}bc$.

Second inequality. Let $U = \sum_k T_k$. Since $U = AU$, Cauchy–Schwarz gives $(\mathbb{E}U)^2 \leq \mathbb{E}A \cdot \mathbb{E}U^2$, and $U^2 = \sum_k T_k + \sum_{k \neq \ell} T_k T_\ell$ with $T_k T_\ell \leq B_k C_\ell$. Hence $(tz)^2 \leq a[tz + t(t - 1)bc]$, which rearranges to $t(z^2 - abc) \leq az - abc$.

Rejecting order. If $z^2 > abc$, the second inequality fails as soon as $t > (az - abc)/(z^2 - abc)$; the fan can be rooted at whichever of A, B, C has the smallest zero marginal, giving (9). The ratio is at least 1 since numerator minus denominator equals $z(\min\{a, b, c\} - z) \geq 0$, so the bound is never below 2, consistent with order one never rejecting. The smaller feasible sets of (1) reject no later. $\square$

Inequality (11) is a pointwise marginal-polytope certificate in the sense of [11], on a $1 \times 1 \times t$ box of source copies; here it is available in closed form at every order.

Proof of Proposition 4.2. R_p has $a = b = c = z = p$, so $z^2 = p^2 > p^3 = abc$ and Lemma 3.7 gives incompatibility. Order one is passed by every law. At order two the first fan inequality reads $2p \leq p + p^2$, which fails for $p < 1$; so $t_{\min}^H(R_p) = 2$ for all three hierarchies, while $d_{\text{TV}}(R_p, \mathcal{C}_\Delta) \leq p \rightarrow 0$. $\square$

5 Sharp rejecting-order exponent

5.1 Finite rejection for each incompatible law

Navascués and Wolfe [7] prove that for every correlation scenario G and every $P \in \mathcal{I}_t^{\text{NW}}(G)$ there is a compatible law within Euclidean distance $O(t^{-1/2})$ of P , with an explicit constant depending on the number of source types; since the compatible set is closed [9], every incompatible P is rejected at some finite order:

$$P \notin \mathcal{C}_G \implies \exists t : P \notin \mathcal{I}_t^{\text{NW}}(G).$$

We recall the precise finite de Finetti estimate behind this in the proof of Proposition 7.1.

5.2 The family P_ε

Proposition 5.1 (One family with divergent rejecting order). *For $0 < \varepsilon < 1/8$ let $P_\varepsilon = Q(\varepsilon, 1 - \frac{1}{2}\varepsilon^{2/3})$, and write $\sigma = \frac{1}{2}\varepsilon^{2/3}$, $m = \varepsilon + (1 - \varepsilon)\sigma$ for the common one-variable zero marginal and $z = \varepsilon + (1 - \varepsilon)\sigma^3$ for $P_\varepsilon(000)$. Then $P_\varepsilon \notin \mathcal{C}_\Delta$, $P_\varepsilon \rightarrow \delta_{111} \in \mathcal{C}_\Delta$ as $\varepsilon \downarrow 0$, and for $H \in \{\text{exp}, \text{AI}, \text{NW}\}$ the first rejecting order $t_{\min}^H(P_\varepsilon) = \min\{t : P_\varepsilon \notin \mathcal{I}_t^H\}$ is finite and satisfies*

$$\left\lfloor 1 + \frac{1}{2}\varepsilon^{-1/3} \right\rfloor + 1 \leq t_{\min}^H(P_\varepsilon) \leq \lfloor \tau_-(\varepsilon) \rfloor + 1 \quad (\varepsilon \leq 1/64),$$

$$\tau_- = \frac{z + m^2/2 - \sqrt{(z + m^2/2)^2 - 2m^3}}{m^2},$$

so that

$$\frac{1}{2} \leq \liminf_{\varepsilon \downarrow 0} \varepsilon^{1/3} t_{\min}^H(P_\varepsilon) \leq \limsup_{\varepsilon \downarrow 0} \varepsilon^{1/3} t_{\min}^H(P_\varepsilon) \leq 4 - 2\sqrt{3} \approx 0.536.$$

In particular $t_{\min}^H(P_\varepsilon) = \Theta(\varepsilon^{-1/3})$ for each of the three hierarchies.

The lower bound is the passing construction of Section 3 and the upper bound is the first fan inequality of Theorem 4.1; they agree in exponent and not in constant. Section 6 proves the distance statements for the same family.

5.3 Proof of the exponent bounds

Proof of Proposition 5.1. Put $\sigma = \frac{1}{2}\varepsilon^{2/3}$, so $P_\varepsilon = Q(\varepsilon, 1-\sigma)$, $m = \varepsilon + (1-\varepsilon)\sigma$ and $z = \varepsilon + (1-\varepsilon)\sigma^3$. *Incompatibility:* $z \geq \varepsilon$ and $m \leq \varepsilon + \sigma = \varepsilon^{2/3}(\varepsilon^{1/3} + \frac{1}{2})$, so $m^3 < \varepsilon^2 \leq z^2$ because $\varepsilon^{1/3} < \frac{1}{2}$ when $\varepsilon < 1/8$; Lemma 3.7 applies. *Limit:* $\varepsilon \rightarrow 0$ and $1 - \sigma \rightarrow 1$, so $P_\varepsilon \rightarrow \delta_{111}$, which is compatible. *Lower bound:* if $t \leq 1 + \frac{1}{2}\varepsilon^{-1/3}$ then $(t-1)\varepsilon \leq \sigma$, and Bernoulli's inequality gives $(1-\varepsilon)^{t-1} \geq 1 - (t-1)\varepsilon \geq 1 - \sigma$, so Theorem 3.1 applies with $r = 1 - \sigma$ and $P_\varepsilon \in \mathcal{I}_t^{\text{exp}}$; hence every rejecting order exceeds $\lfloor 1 + \frac{1}{2}\varepsilon^{-1/3} \rfloor$. Finiteness is asymptotic completeness together with (1).

Upper bound. Write $v_t = tz - m - \binom{t}{2}m^2$; by the first fan inequality with $a = b = c = m$, positivity of v_t rejects at order t . For $t = x\varepsilon^{-1/3} + O(1)$ one has $tz = x\varepsilon^{2/3} + O(\varepsilon)$, $m = \frac{1}{2}\varepsilon^{2/3} + O(\varepsilon)$ and $\binom{t}{2}m^2 = \frac{x^2}{8}\varepsilon^{2/3} + O(\varepsilon)$, so

$$v_t = \left(x - \frac{1}{2} - \frac{x^2}{8}\right)\varepsilon^{2/3} + O(\varepsilon),$$

whose leading coefficient is positive exactly for $4 - 2\sqrt{3} < x < 4 + 2\sqrt{3}$. Fixing any x slightly above $4 - 2\sqrt{3}$ gives rejection for all small ε , which is the limsup bound. For the exact finite statement, v_t is the quadratic $-\frac{m^2}{2}t^2 + (z + \frac{m^2}{2})t - m$ in t , with discriminant $D = (z + m^2/2)^2 - 2m^3$ and roots $\tau_\pm = (z + m^2/2 \pm \sqrt{D})/m^2$. For $\varepsilon \leq 1/64$ one has $m \leq \frac{3}{4}\varepsilon^{2/3}$ and $z \geq \varepsilon$, so $D \geq \varepsilon^2 - 2m^3 \geq \varepsilon^2(1 - \frac{27}{32}) = \frac{5}{32}\varepsilon^2$ and $\tau_+ - \tau_- = 2\sqrt{D}/m^2 \geq \frac{4\sqrt{10}}{9}\varepsilon^{-1/3} > 1$; since $v_1 = z - m < 0$ the integer $\lfloor \tau_- \rfloor + 1$ lies strictly between the roots, where $v_t > 0$. Hence $t_{\min}^{\text{NW}}(P_\varepsilon) \leq \lfloor \tau_- \rfloor + 1 = (4 - 2\sqrt{3})\varepsilon^{-1/3} + O(1)$. □

The gap between the constants $\frac{1}{2}$ and $4 - 2\sqrt{3}$ is a gap between the passing construction and the fan certificate; whether $\varepsilon^{1/3}t_{\min}^H(P_\varepsilon)$ converges, and to what, is open. Exact certificates at $\varepsilon = k^{-3}$ show that the two bounds coincide for $k \in \{4, 6, 8, 10, 16\}$ (first rejecting order 4, 5, 6, 7, 10) and differ at the tested values $k \geq 32$ (Section 11).

6 Distance to the compatible set

The parameter ε of the family P_ε is, to leading order, a fixed multiple of the distance from P_ε to the compatible set, in total variation and in Euclidean norm. Combined with Proposition 5.1 this turns the exponent into a statement about geometry near the boundary of $\mathcal{C}_\Delta$.

Theorem 6.1 (Distance asymptotics). *For $P_\varepsilon = Q(\varepsilon, 1 - \frac{1}{2}\varepsilon^{2/3})$, $0 < \varepsilon < 1/8$,*

$$d_{\text{TV}}(P_\varepsilon, \mathcal{C}_\Delta) = (1 - 2^{-3/2})\varepsilon + O(\varepsilon^{4/3}), \quad d_2(P_\varepsilon, \mathcal{C}_\Delta) = \sqrt{8/7}(1 - 2^{-3/2})\varepsilon + O(\varepsilon^{4/3}).$$

Corollary 6.2 (Direction-specific boundary complexity). *For $H \in \{\text{NW}, \text{AI}, \text{exp}\}$,*

$$t_{\min}^H(P_\varepsilon) = \Theta(d_{\text{TV}}(P_\varepsilon, \mathcal{C}_\Delta)^{-1/3}) = \Theta(d_2(P_\varepsilon, \mathcal{C}_\Delta)^{-1/3}) \quad (\varepsilon \downarrow 0).$$

This is a statement about one direction of approach to the boundary; by Proposition 4.2 no such lower bound holds uniformly over all incompatible laws.

Proof of Theorem 6.1. Put $\sigma = \frac{1}{2}\varepsilon^{2/3}$, $m = \varepsilon + (1 - \varepsilon)\sigma$, $z = \varepsilon + (1 - \varepsilon)\sigma^3$, $c_0 = 2^{-3/2}$ and $d_0 = 1 - c_0$. We prove the lower bounds and then construct one compatible law attaining both leading constants.

Lower bounds. The product law $\text{Bern}(1 - \sigma)^{\otimes 3}$ is compatible and within total variation ε of P_ε . Compactness of $\mathcal{C}_\Delta$ gives a nearest law in either norm [9]. A total-variation minimizer Q therefore has distance $\delta = O(\varepsilon)$. Since probabilities of events differ by at most δ ,

$$Q(000) \geq z - \delta, \quad Q_V(0) \leq m + \delta \quad (V = A, B, C).$$

Finner gives

$$z - \delta \leq (m + \delta)^{3/2} = c_0\varepsilon + O(\varepsilon^{4/3}),$$

so $\delta \geq d_0\varepsilon - O(\varepsilon^{4/3})$.

A Euclidean minimizer Q_2 also satisfies $\|P_\varepsilon - Q_2\|_2 = O(\varepsilon)$, by comparison with the product law. Cauchy–Schwarz bounds each one-variable marginal error by $2\|P_\varepsilon - Q_2\|_2$, since each marginal sums four atoms. Hence Finner gives $Q_2(000) \leq c_0\varepsilon + O(\varepsilon^{4/3})$. Set $D = P_\varepsilon(000) - Q_2(000)$; then $D \geq d_0\varepsilon - O(\varepsilon^{4/3}) > 0$ for small ε . The other seven coordinates of $Q_2 - P_\varepsilon$ sum to D , so

$$\|Q_2 - P_\varepsilon\|_2^2 \geq D^2 + \frac{D^2}{7}.$$

This proves the Euclidean lower bound.

A compatible approximation. Set $h = d_0/7$ and $u = \sigma + (c_0 + h)\varepsilon$. Each source carries a base bit with success probability $\sqrt{u}$ and a flag with success probability $h\varepsilon$, all six bits independent. A node outputs 0 if both incident base bits equal 1 or either incident flag equals 1. These probabilities lie in $[0, 1]$ for small ε , and the resulting law Q_ε belongs to $\mathcal{C}_\Delta$.

Without flags, the atom probabilities are $u^{3/2}$ at 000, $u - u^{3/2}$ at each one-zero atom, zero at each two-zero atom, and $1 - 3u + 2u^{3/2}$ at 111. A single flag forces the two nodes incident to its source to zero. Except on a base-output event of probability $O(u)$, it transfers mass from 111 to the two-zero atom associated with that source. Multiple flags have probability $O(\varepsilon^2)$. The errors from these exceptions are $O(\varepsilon u + \varepsilon^2) = O(\varepsilon^{5/3})$.

Using $u^{3/2} = c_0\varepsilon + O(\varepsilon^{4/3})$ gives the following atom probabilities, with an $O(\varepsilon^{4/3})$ remainder in each entry:

Atom	P_ε	Q_ε
000	ε	$c_0\varepsilon$
Each of 011, 101, 110	σ	$\sigma + h\varepsilon$
Each of 001, 010, 100	0	$h\varepsilon$
111	$1 - 3\sigma - \varepsilon$	$1 - 3\sigma - (c_0 + 6h)\varepsilon$

Since $1 - c_0 = 7h$, the deviation at 000 is $-d_0\varepsilon + O(\varepsilon^{4/3})$ and at each of the other seven atoms it is $h\varepsilon + O(\varepsilon^{4/3})$. Its total variation is $d_0\varepsilon + O(\varepsilon^{4/3})$ and its Euclidean norm is $\sqrt{8/7}d_0\varepsilon + O(\varepsilon^{4/3})$, proving both upper bounds. $\square$

Corollary 6.2 follows from Theorem 6.1 and Proposition 5.1. The constant $1 - 2^{-3/2}$ is the fraction of the rare atom $P_\varepsilon(000) \approx \varepsilon$ that no compatible law can retain once its zero marginals are pinned near $\frac{1}{2}\varepsilon^{2/3}$, by the Finner inequality.

7 General distance-promised rejecting-order bounds

Theorem 3.2 says that the order needed cannot be fixed in advance for the scenario. Whether it can be bounded in advance for the instance is a different question. Proposition 7.1 bounds it under a distance promise, and Proposition 8.1 bounds it in the input size.

Proposition 7.1 (Distance-promised rejecting order). *Let G be a correlation scenario with L independent source types (private response randomness may be absorbed into incident sources), in which every observed node has a latent parent, with joint observed alphabet of size K , and let $\delta_2 = d_2(P, \mathcal{C}_G)$ and $\delta = d_{\text{TV}}(P, \mathcal{C}_G)$. If $P \in \mathcal{I}_n^{\text{NW}}(G)$ then*

$$\delta_2^2 \leq \left[1 - \left(1 - \frac{1}{n}\right)^L\right] (1 - \|P\|_2^2) \leq \frac{L(1 - \|P\|_2^2)}{n}. \quad (12)$$

Consequently, for incompatible P ,

$$t_{\min}^{\text{NW}}(P) \leq \left\lfloor \frac{L(1 - \|P\|_2^2)}{\delta_2^2} \right\rfloor + 1 \leq \left\lfloor \frac{LK}{4\delta^2} \right\rfloor + 1;$$

for the binary triangle, $t_{\min}^{\text{NW}}(P) \leq \lfloor 6/\delta^2 \rfloor + 1$.

The proof follows the finite-sampling argument of [7], retaining the collision probability and the factor $1 - \|P\|_2^2$. A source-free stochastic node can be given its own independent source, included in L .

Proof of Proposition 7.1. Let Γ be an order- n feasible law for P . Assume first that $n \geq 2$. For a deterministic assignment ω of all copied observations, sample one index uniformly and independently for each source type and read off the observed tuple; this is a genuine model of G with the sampled indices as sources and the entries of ω as deterministic responses, so its output law q_ω is compatible. Averaging over $\omega \sim \Gamma$ gives $\mathbb{E}q_\omega = P$, by symmetry and the degree-one diagonal condition. Two independent samples of the index tuples collide in no source type with probability $\alpha = (1 - 1/n)^L$, and conditionally on no collision the pair is, by symmetry and the degree-two diagonal condition, distributed as $P \otimes P$; so $\mathbb{E}q_\omega^{\otimes 2} = \alpha P^{\otimes 2} + (1 - \alpha)R$ for some law R . Since $\mathbb{E}q_\omega = P$,

$$\mathbb{E}\|q_\omega - P\|_2^2 = \mathbb{E}\left[\sum_x q_\omega(x)^2\right] - \|P\|_2^2 = (1 - \alpha)(R\{x = y\} - \|P\|_2^2) \leq (1 - \alpha)(1 - \|P\|_2^2).$$

For $n = 1$, the law q_ω is a point mass and $\mathbb{E}\|q_\omega - P\|_2^2 = 1 - \|P\|_2^2$, which gives the same bound. Some ω attains at most the mean, and q_ω is compatible, which proves (12); the last inequality there is $1 - (1 - 1/n)^L \leq L/n$. For incompatible P , feasibility at order n forces $n \leq L(1 - \|P\|_2^2)/\delta_2^2$, and $\|P - Q\|_2^2 \geq \|P - Q\|_1^2/K = 4d_{\text{TV}}(P, Q)^2/K$ gives the second form; the triangle has $L = 3$ and $K = 8$. This is the argument of [7, Section 4.2 and eq. (A7)] with the collision probability kept exact and a single compatible component extracted from the mixture in place of a convex combination; no convexity of $\mathcal{C}_G$ is used. Deterministic responses suffice because the sampled tables are deterministic, so L counts the sources of the scenario without private response roots. $\square$

For the binary triangle, the worst-case distance

$$H_n = \sup_{P \in \mathcal{I}_n^{\text{NW}}} d_{\text{TV}}(P, \mathcal{C}_\Delta)$$

satisfies $H_n = \Omega(n^{-3})$ by Proposition 5.1 and Theorem 6.1, taking $\varepsilon = [2(n - 1)]^{-3}$ for large n , and $H_n = O(n^{-1/2})$ by (12). The sharp rate remains open.

8 Rational input bit-length complexity

Proposition 8.1 (Required order in the input size). *Encode each of the eight rational atoms by its reduced numerator and positive denominator in ordinary binary, with delimiters of linear overhead. For a law P on three bits, let $\ell(P)$ be its total bit length. For each $H \in \{\text{NW}, \text{AI}, \text{exp}\}$,*

$$\max\{t_{\min}^H(P) : \ell(P) \leq B, P \notin \mathcal{C}_\Delta\} = 2^{\Theta(B)}.$$

Proof of Proposition 8.1. Lower bound. With $k = 2^j$, $\varepsilon = k^{-3}$ and $\sigma = 1/(2k^2)$, the eight atoms of P_ε are rationals with denominator $8k^9$ and numerators of $O(j)$ bits, so $\ell(P_\varepsilon) = O(j)$, while Proposition 5.1 gives $t_{\min}^H(P_\varepsilon) > k/2 = 2^{j-1}$. Choosing j proportional to B gives $2^{\Omega(B)}$.

Upper bound. By [9], $\mathcal{C}_\Delta$ is the image of a compact semialgebraic parameter space (six latent values per source and stochastic binary response tables) under a polynomial map, hence a closed semialgebraic subset of the simplex. The function $u(P) = d_2(P, \mathcal{C}_\Delta)^2$ is therefore semialgebraic, and its graph is described by a fixed quantifier-free formula $\Phi(P, u)$ in integer polynomials of fixed degrees, by real quantifier elimination [1]. For a rational P of bit length B , substituting P and clearing denominators turns the atoms of Φ into univariate integer polynomials in u of fixed degree and coefficient height $2^{O(B)}$. The set of u satisfying $\Phi(P, u)$ is the single point $u(P)$; discard identically zero specialized polynomials, whose comparisons have constant truth values. If none of the remaining polynomials vanished at $u(P)$, their signs would persist on a neighbourhood and the formula would define an interval. Thus some nonzero specialized polynomial f vanishes at $u(P)$. Remove any factor u^r from f to obtain $g(u) = a_0 + \dots + a_d u^d$ with $a_0 \neq 0$, the same nonzero root, and coefficient height at most $H = 2^{O(B)}$. For $0 < u < 1/(H+1)$,

$$\left| \sum_{i=1}^d a_i u^i \right| \leq H \sum_{i=1}^d u^i < \frac{Hu}{1-u} < 1 \leq |a_0|,$$

so $g(u) \neq 0$. Hence $u(P) \geq 2^{-O(B)}$ when P is incompatible, and Proposition 7.1 gives $t_{\min}^{\text{NW}}(P) \leq 3(1 - \|P\|_2^2)/u(P) + 1 = 2^{O(B)}$; the two smaller feasible sets reject no later. $\square$

For the fixed binary triangle, a quantifier-free membership formula contains a fixed number of fixed-degree integer polynomials. Evaluating their signs at a rational input takes time polynomial in B . Quantifier elimination supplies such a formula in principle [1], although deriving and storing it may be impractical. Thus the exponential inflation-order bound coexists with a polynomial-time decision procedure for this fixed scenario. For an incompatible rational input, one can also compute the first rejecting order by enumerating exact rational feasibility problems.

9 Relation to prior work

Finite-order convergence. Navascués and Wolfe [7, Section 4.1] give a binary family that passes second-order triangle inflation while violating Finner, and ask whether any finite order characterizes compatibility. Theorem 3.2 extends that separation to every order with a different family. They also exhibit a polynomial optimization problem whose inflation relaxations fail to attain the exact optimum at any finite order, in a scenario where every law is compatible. Our result concerns the compatibility tests themselves.

Network inequalities. Finner’s inequality [3] supplies the incompatibility test. Renou et al. [8] prove its triangle-network form for quantum sources and for no-signalling boxes combined by local wirings, and give the classical Cauchy–Schwarz argument used here. The explicit rational violation margin in Lemma 3.8 comes from our choice of parameters.

The first fan inequality follows the marginal-polytope method of Wolfe, Spekkens and Fritz [11, Section IV.D]: prove a pointwise inequality on copied observations, then substitute the prescribed marginals. Fraser and Wolfe [4] derive triangle inequalities from the wagon-wheel and web inflations. Theorem 4.1 gives a closed formula indexed by the order, together with a second-moment bound. Along P_ε , its rejecting order matches the exponent of the passing construction. Recent work of da Silva, Pozas-Kerstjens and Parisio [2] studies local models and conjectural boundaries for permutation-invariant binary-triangle distributions; our bounds concern how the inflation order grows near a compatible point.

Quantitative convergence and effectivity. Proposition 7.1 refines the finite-sampling estimate in [7] by retaining the exact collision probability and the factor $1 - \|P\|_2^2$. The bit-length upper bound combines this estimate with the finite latent-cardinality theorem of Rosset, Gisin and Wolfe [9], quantifier elimination, and an elementary bound on nonzero polynomial roots. Its lower bound uses rational members of P_ε .

Quantum inflation. Wolfe et al. [10] develop quantum inflation, and Ligthart, Gachechiladze and Gross [6] construct a provably convergent variant. The distinction between quantum models matters for such convergence statements. The present nontermination theorem applies to classical inflation and the two strengthenings defined in Section 2.

10 Open problems

The construction and the fan inequality explain the exponent $1/3$ through the same scales. Along P_ε , the all-zero atom has probability of order ε , while each zero marginal has order $\varepsilon^{2/3}$. In the defect cube, the $t - 1$ other cells on each line contribute at most $(t - 1)\varepsilon$ to that marginal. The construction therefore passes when t is of order $\varepsilon^{-1/3}$. In a fan of that size, both tz and $m + \binom{t}{2}m^2$ have order $\varepsilon^{2/3}$. Their leading coefficients give the upper constant $4 - 2\sqrt{3}$.

Question 10.1 (Exact leading constant). Does $\varepsilon^{1/3}t_{\min}^H(P_\varepsilon)$ converge? If so, what is its limit, and do the three hierarchies share it?

The current bounds give the interval $[\frac{1}{2}, 4 - 2\sqrt{3}]$. The finite certificates determine the first rejecting order at $\varepsilon = k^{-3}$ for $k \in \{4, 6, 8, 10, 16\}$; the passing and fan bounds differ at the larger tested values.

Question 10.2 (Worst-case convergence exponent). For $H_n = \sup_{P \in \mathcal{I}_n^{\text{NW}}} d_{\text{TV}}(P, \mathcal{C}_\Delta)$, which rate between the bounds $\Omega(n^{-3})$ and $O(n^{-1/2})$ is sharp?

Question 10.3 (Explicit rejecting orders beyond Finner). Can one bound the rejecting order in directly evaluable quantities of an incompatible input law that satisfies every Finner inequality?

Theorem 4.1 covers Finner violations. The distance and bit-length bounds apply more broadly, but computing the distance or extracting the constants from quantifier elimination need not be practical.

Question 10.4 (Finite stabilization). Which correlation scenarios and observed alphabet sizes admit a finite characterizing order?

The star scenarios of [7] terminate at order two. The binary triangle, and hence every triangle with at least two outcomes at each node, does not. The defect construction uses the fact that each pair of observed nodes shares a source; extending it to other graphs requires a corresponding construction of the copied marginals.

Question 10.5 (Quantum and no-signalling analogues). For which quantum or no-signalling inflation hierarchies can one construct incompatible laws with an unbounded first rejecting order and determine its growth?

This question requires specifying both the hierarchy and the compatible resource model; the classical proof here does not decide it.

11 Reproducibility

The proofs do not depend on computation. The accompanying artifact contains exact rational certificates and checkers for the following finite instances:

- structural certificates at orders $t = 1, 2, 3$: the root law, the line supports (5), and the target atoms of P_t as exact rationals; the checker verifies nonnegativity and normalization, covariance of the supports under every adjacent-transposition generator of S_t^3 , all t^3 copied-triangle marginals by exact enumeration over the relevant roots, mutual disjointness of the diagonal supports R_ℓ , all 8^t diagonal equations against $P_t^{\otimes t}$, and the exact Finner gap;
- a sparse full order-two inflation table with 166 nonzero coordinates out of 2^{12} and denominator 2^{32} , produced by a separate enumeration of all 256 defect configurations; the checker treats it as an untrusted linear-programming vector and verifies normalization, nonnegativity, all 12,288 generator-coordinate symmetry equations and all 64 diagonal equations;
- a negative control: the same table with one denominator unit moved from the all-one to the all-zero atom, which is normalized and nonnegative and is rejected by the diagonal equations;
- an exact rational order-two Farkas certificate rejecting $R_{1/2}$ from Proposition 4.2;
- fan certificates: for fourteen members P_ε , $\varepsilon = k^{-3}$, the exact atoms, the Bernoulli passing certificate at order $1 + k/2$, the exact fan value v_t at the first rejecting order of the fan inequality, and the second-moment rejecting order; an all-order certificate of the pointwise inequality (11) as a nonnegative polynomial identity in the count S , checked exhaustively over 174,760 assignments and 271,500 count orbits; the 2,660 index-permutation embeddings behind (10); and eight rejected negative controls. The checker uses explicit exceptions and passes identically under an optimized interpreter.

The certified first rejecting orders of P_ε for $\varepsilon = k^{-3}$ are 4, 5, 6, 7, 10 at $k = 4, 6, 8, 10, 16$, where the passing and fan bounds meet; for $k = 32, 128, 1024, 65536$ the fan bounds are 19, 70, 551, 35122 against lower bounds on the first rejecting order of 18, 66, 514, 32770. The exact Finner gaps at $t = 1, 2, 3$ are $1/8$, $9273825/2^{32}$ and $327279338388988503553/1792720717930698493280256$, above the uniform bounds $\varepsilon_t^2/2 = 1/8$, $1/512$, $1/5832$. The certificate checkers use Python integers and exact fractions, without a linear-programming solver or floating-point comparisons. The hardened nontermination checker raises an explicit exception on every failed check and refuses to run under an optimized interpreter; the artifact keeps the original checker unchanged for provenance

and includes mutation tests (truncated, empty, corrupted, wrong-order and unsymmetric inputs, and optimized-interpreter runs) with their logs. A manifest with SHA-256 identities and a fresh replay record with operating system, interpreter version, commands, wall times and output hashes accompany the artifact.

To replay the checks, use Python 3 from the directory containing `paper/` and `artifact/`:

```
python3 -B artifact/verifiers/run_replay.py
python3 -B artifact/verifiers/mutation_tests.py
python3 -B artifact/verifiers/mutation_tests_fan.py
```

The replay runner copies each certificate set into a temporary directory and writes new records under `artifact/replay_logs/`. Its independent distance check requires `mpmath`; the exact certificate checkers use the Python standard library. Replaying changes the log files, so the supplied manifest identifies the distributed snapshot, not regenerated logs. The finite checks verify the listed tables and certificates; the all-order and distance-asymptotic claims rest on the proofs above.

Formal verification. Theorem 3.2 and the results it rests on are formalized in Lean 4 with Mathlib: the finite-alphabet form of Lemma 3.7; the defect-cube law with its symmetry, diagonal law, injectable marginals and ancestral products (Lemmas 3.4, 3.5, 3.6 and 3.9, and Theorem 3.1 for $\mathcal{I}_t^{\text{AI}} \subseteq \mathcal{I}_t^{\text{NW}}$); the margin of Lemma 3.8; Appendix A; Theorem 4.1 with the rejecting-order bound; Proposition 4.2; the finite bounds of Proposition 5.1; and the triangle case of Proposition 7.1. The headline statement is `TriangleInflation.no_finite_characterizing_order`. The development contains no `sorry`, and every theorem depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. Three boundaries are recorded with the sources. The recursively expressible hierarchy of Definition 2.3 is not formalized, so the formal membership statements cover the two larger feasible sets. The formal compatible set uses finite latent alphabets, which by [9] loses nothing but is not itself formalized, so the formal incompatibility statements are the weaker reading. Theorem 6.1, Proposition 8.1 and the limit form of Proposition 5.1 are not formalized. The Lean sources accompany the artifact.

A Injectable sets of the triangle

We justify the description of injectable sets used in Definition 2.2. In the order- t inflated triangle, a set S of copied observations is injectable when the map that erases copy indices is injective on S and, for any two members of S that are copies of variables sharing a source in the triangle, the copy index of that source agrees [11, Definition 4]. Injectivity allows at most one copy of each of A , B , C . If S contains A^{ij} and $B^{i'k}$ then $i = i'$; if it contains A^{ij} and $C^{j'k}$ then $j = j'$; if it contains B^{ik} and $C^{jk'}$ then $k = k'$. Hence $S \subseteq \Delta_{ijk}$ for suitable (i, j, k) , any missing index being chosen freely. Conversely every subset of a copied triangle is injectable. This proves the characterization for all t .

The degree- t diagonal condition (ii) of Definition 2.1 also implies the lower-degree conditions of [7]: the law of any $g \leq t$ diagonal triangles is a marginal of $P^{\otimes t}$, hence $P^{\otimes g}$, and by symmetry the same holds for any g copied triangles with pairwise distinct indices in all three families.

B Certificate formats

Structural certificate at order t . A JSON object with the root law (kind, $\varepsilon_t = 1/(2t^3)$ and $1 - \varepsilon_t$ as exact fractions), the value r_t , the list of t^3 roots D_{ijk} , for each of the $3t^2$ outputs the list of roots on its line, the eight target atoms of P_t as exact fractions, and the claimed counts of diagonal equations (8^t) and symmetry generators ($3(t - 1)$). The checker recomputes every claimed quantity from the definitions and compares exactly. The checker computes the copied-triangle marginals by enumerating the 2^{3t-2} assignments of the roots on the three lines through a cell, which makes $t = 3$ (with 2^7 assignments per triangle and $8^3 = 512$ diagonal equations) immediate.

Full order-two table. A JSON object listing the 12 observation names in a fixed order, the denominator $2^{32} = (2t^3)^{t^3}$ at $t = 2$, and the nonzero numerators $\Gamma_2(w)$ of (6) for 166 assignments $w \in \{0, 1\}^{12}$; unlisted assignments have numerator 0. The checker verifies $\sum_w \Gamma_2(w) = 1$, nonnegativity, $\Gamma_2(w) = \Gamma_2(\pi w)$ for the three adjacent transpositions π and all 4096 assignments, and the 64 equations $\sum_{w: w_{\text{diag}}=(x,y)} \Gamma_2(w) = P_2(x)P_2(y)$ for the diagonal coordinates $(A^{11}, B^{11}, C^{11}, A^{22}, B^{22}, C^{22})$.

Negative control. The same object with the numerators of the all-one and all-zero assignments changed by -1 and $+1$. Normalization and nonnegativity hold; the diagonal equation at $(000, 000)$ fails. The checker is required to reject it.

Promised-rejection calibration. An exact Farkas certificate showing that the rational law $\frac{1}{2}(\delta_{000} + \delta_{111})$ is infeasible at order two with margin $1/2$, which checks rejection of $R_{1/2}$ from Proposition 4.2. This finite check does not certify the general convergence constants.

Fan certificates. A JSON list of records, one per k in the list 4, 6, 8, 10, 16, 32, 64, 128, 256, 512, 1024, 4096, 16384, 65536, with $\varepsilon = k^{-3}$, $\sigma = 1/(2k^2)$, the eight exact atoms of P_ε , the values m , z and the Finner gap $z^2 - m^3$, the passing order $1 + k/2$ with the exact Bernoulli certificate $(t - 1)\varepsilon \leq \sigma$, the first order at which $v_t = tz - m - \binom{t}{2}m^2$ is positive together with the exact rational value of v_t there, and the first order at which the second-moment inequality fails. An all-order certificate records the identity $1 + \frac{1}{2}S(S - 1) - S = \frac{1}{2}(S - 1)(S - 2)$ and its nonnegativity on integers by cases, so that (11) needs no order-by-order enumeration; the checker nevertheless enumerates all assignments of the fan for small t and all count orbits for larger t , and verifies that the index permutations used in (10) extend to permutations of $[t]$. Negative controls perturb the atoms, the parameters and the claimed orders and must be rejected.

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